Polymer Films for Retinal Prostheses: Material Properties and Cellular Response

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Retinal degeneration, associated with aging and metabolic disorders as diabetes, leads to photoreceptor loss and visual impairment, including blindness. Retinal prostheses offer a promising strategy to functionally replace damaged photoreceptors and partially restore vision. In this work, we investigated polymer-based semiconducting films as candidate materials for subretinal interfaces. Blends were composed of PEDOT:PSS combined with TQ1 or RP3HT to achieve high photosensitivity, efficient charge transport, and cytocompatibility.

Films with varying compositions were prepared by spin-casting. Chemical composition was characterized by Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS) and surface morphology by Atomic Force Microscopy (AFM). Photosensitivity and light-dependent polymer–cell interactions were studied under controlled irradiation cycles. Cytocompatibility of the blends with ARPE-19 cells was assessed through LDH assay, fluorescent staining, and reactive oxygen species (ROS) detection, both in darkness and under illumination.

ToF-SIMS and AFM measurements revealed uniform film structure with both constituent polymers present at the surface. Blends showed photosensitive properties, higher for HTL:TQ1. The films showed no cytotoxicity in darkness, however, lowered viability of the cells, together with ROS generation was observed after illumination for the cells cultured on HTL:TQ1 films. Therefore, we investigated blends with varying TQ1 abundance to identify compositions that exhibit light-induced electrical activity while remaining non-cytotoxic.

The obtained results demonstrate the potential of semiconducting polymer blend films as functional materials for subretinal prosthetic interfaces. They showed combined optoelectronic performance, stability under aqueous environment, and compatibility with ARPE-19 cells, which demonstrates their potential for the development of light-responsive retinal implants.

The study was funded by the “Research support module” as part of the “Excellence Initiative - Research University” program at the Jagiellonian University in Kraków the (A.C, RSM/80/CA).